The Ecology of Deserts

In **desert ecology** is the sum of the interactions between both biotic and abiotic factors in a desert habitat, ecosystem, and community. Some of the abiotic factors also include latitude and distance from the coast, soil, and climate.

Deserts are most notable for their dry climates resulting from rain-blocking mountain ranges and/or remoteness from oceanic moisture. Deserts occupy one-fifth of the Earth's land surface and occur in two belts: between 15° and 35° latitude in both the southern and northern hemispheres. This can be seen in the diagram below. D**eserts** receive an extremely low amount of precipitation, less than enough to support growth of most plants. Deserts are defined as areas with an average annual precipitation of less than 250 ml per year, Perth has a rainfall of approx 852ml per year. Hot deserts usually have a large diurnal (day) and seasonal temperature range, with high daytime temperatures (45 °C), and low night time temperatures (0 °C). During daylight most of the sun's heat reaches the ground (no clouds), and as soon as the sun sets the desert cools quickly by radiating its heat into space.



Deserts support diverse communities of plant and animals that have evolved resistance extreme temperatures and arid conditions. Desert ecology is characterized by dry, alkaline soils, low net production and opportunistic feeding patterns by herbivores and carnivores. Lichens and blue-green algae are significant primary producers in the desert.



Sandy desert



Ice desert

The 10 largest deserts Rank Desert Area

1 Antarctica 13,829,430 km²

2 Arctic 13,700,000 km²

3 Sahara 9,100,000 km²
647,000 km²

4 Arabian Desert 2,330,000 km²

5 Gobi Desert 1,300,000 km²

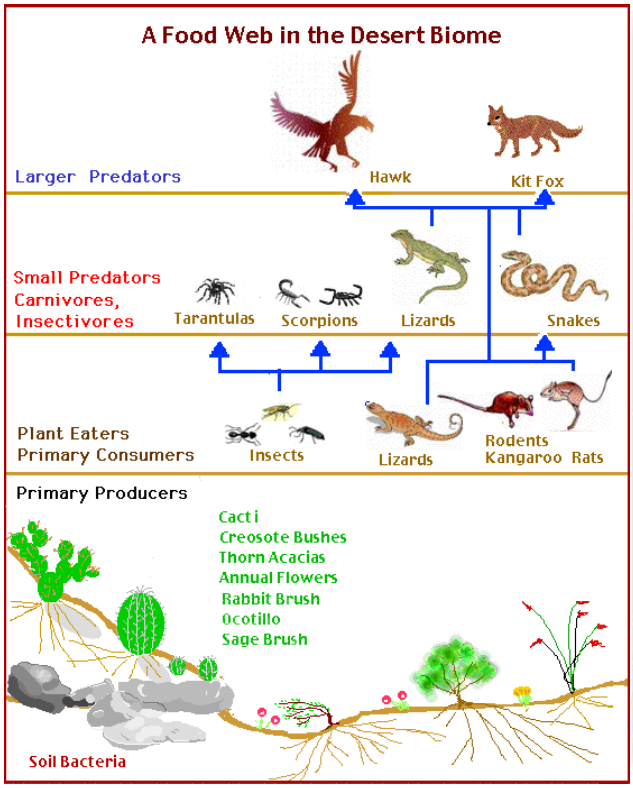
6 Kalahari Desert 900,000 km²

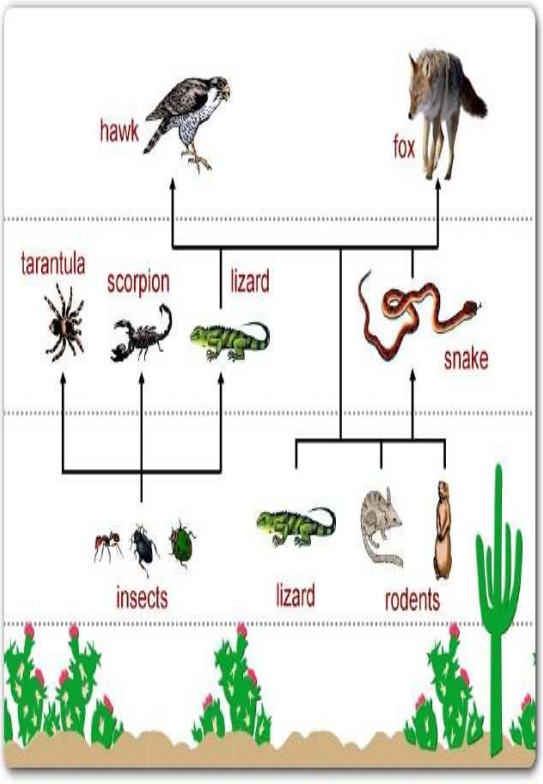
7 Patagonian Desert 670,000 km²

8 Great Victoria Desert (Australia)

9 Syrian Desert 520,000 km²

10 Great Basin Desert 492,000 km²





Typical plants in a desert are cacti, succulents and pine trees. Plants that survive in a low water environment because of their adaptations are called xerophytes. Here are some of the adaptations found in xerophytes.

waxy stomata prickly pear, few stomata, stomata open at night, leaves very small, Stem does photosynthesis, storage of water (succulent leaves), Fleshy tuber, absorbing surface moisture from leaf hairs.

Q1. Why is the artic considered a desert if there is so much ice and snow there?

Q2. Why are deserts so hot during the day?

Q3. Name 5 adaptations animals have to survive in hot deserts.

Q4. Name 5 adaptations plants have to survive in hot deserts.

Q5. Name 5 abiotic characteristics of a typical desert.

Q6. What are xerophytes?

Q7 Suggest several ways a camel is suited to life in the desert?